Daeyoung Kim

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About

I am a Ph.D. student in Computer Science at the University of Tennessee, Knoxville, with a B.S. in EECS from UC Berkeley and a prior career as a professional artist and instructor. My research interests center on human-computer interaction, particularly how interactive systems and AI can support human learning, creativity, and skill development. My background spans computer graphics, HCI research, and teaching, including developing a real-time watercolor simulator and tutoring mathematics and computer science students.

Education

Ph.D. Student in Computer Science Aug 2026 – Present
University of Tennessee, Knoxville – Knoxville, TN *Current Advisor: Dr. Doowon Kim*

- Research interests: human-computer interaction, human-centered computing, computer graphics and interactive systems
- Current research focuses on phishing-email analysis and hidden-text techniques in email previews

B.S. Electrical Engineering & Computer Sciences (EECS) Aug 2023 – May 2026
University of California, Berkeley – Berkeley, CA

- GPA: 3.74 / 4.00
- Focus: computer graphics, human-computer interaction, and visual computing
- Selected coursework: Computer Graphics, Machine Learning, Artificial Intelligence, Computer Security, Computer Networks, Data Structures & Algorithms, Combinatorics, Discrete Mathematics, Probability, Computer Architecture

Lower Division Computer Science (UC Transfer Preparation) Jan 2021 – May 2023
Berkeley City College — Berkeley, CA

- GPA: 4.00 / 4.00
- Completed all lower division CS and math requirements for UC transfer
- Selected coursework: Data Structures & Algorithms, Multivariable Calculus, Linear Algebra, Differential Equations, Introductory Programming, Physics I–II, General Chemistry
- Math & CS Tutor (18 months): supported students in programming, calculus, and linear algebra

Awards & Fellowships

TCE Fellowship Award — Tickle College of Engineering, University of Tennessee, Knoxville Aug 2026

NSF Graduate Research Fellowship Program (GRFP) — Application submitted Oct 2026

Research Experience

Analysis of Hidden Text in Phishing Email Previews — University of Tennessee, Knoxville, Cybersecurity Aug 2026 – Present

- Investigate phishing emails that use **hidden-text/salting** techniques to exploit differences between machine-visible and human-visible content (Prof. **Doowon Kim**; mentor: **Jaehwan Park**)
- Analyze the **Anti-Phishing Working Group (APWG)** email dataset to measure the prevalence of hidden-salt techniques in real-world phishing messages

- Develop visibility criteria for HTML text based on properties such as **font size**, **text/background color**, and other CSS attributes to determine whether content is visible to users
- Parse and compare **email preview text** with rendered message-body content to identify text that appears in previews but is hidden in the body
- Analyze recurring patterns in hidden-text usage to characterize how phishing emails exploit differences between machine-visible and human-visible content

Undergraduate Research Assistant — UC Berkeley, HCI & Social Computing Nov 2025 – Apr 2026

- Contributed to a research project studying how **visual traits** cluster and spread within **TikTok aesthetic communities** (Prof. **Marti Hearst** and Prof. **Björn Hartmann**; mentor: **Samia Menon**)
- Built exploratory modeling workflows to characterize "trait bundles" inside a single aesthetic using **Apriori** frequent itemsets, **NMF**, and topic modeling over binarized feature representations
- Developed and tested computer-vision extraction approaches: trained **YOLO** for object detection, then benchmarked against **GPT vision API** structured labeling and supported a pipeline switch based on quality/interpretability
- Performed data collection by sourcing creators aligned with a target aesthetic and generating frame/screenshot datasets for analysis
- Conducted related-work citation tracking and contributed to ongoing manuscript revision.

Undergraduate Research Assistant — UC Berkeley, HCI & Interactive Systems Nov 2025 – Jan 2026

- Prototyped **programmable/animated food** interactions using a modified pen-plotter platform (Prof. **Eric Paulos**; mentor: **Vivian Chan**)
- Integrated mechanical + motion components (gantry alignment, fixtures, belt tensioning) and evaluated positioning accuracy/repeatability
- Implemented a **Python-to-G-code** toolchain for generating trajectories and timed animations; enabled rapid iteration across experimental conditions
- Conducted precision and material-behavior tests with magnetic food surrogates; iterated control parameters and paths to improve stability
- Communicated interaction concepts via sketches and polished figures for team discussion and potential publication

Publications

Modeling the Diffusion of Visual Culture on TikTok (manuscript in revision) 2026

- Samia Menon, **Daeyoung Kim**, Björn Hartmann, Marti Hearst.
- Previously submitted to *International AAAI Conference on Web and Social Media (ICWSM 2026)*; manuscript currently being revised for resubmission.
- Contributions: visual-trait clustering experiments (Apriori, NMF, topic modeling), CV baseline (YOLO) prototyping and evaluation, creator sourcing + frame dataset collection, and related-work/citation discovery.

Projects

Digital Watercolor Painting Simulator — Godot, GLSL, GPU Physics 2025 – Present

- Built a GPU accelerated watercolor painting application in Godot Engine 4.4 that simulates realistic water flow, pigment diffusion, evaporation, and deposition in real time on a 256×256 canvas
- Implemented the physics pipeline entirely on the GPU using GLSL compute shaders and Godot's `RenderingDevice` API with double buffered textures for water, mobile pigment, and static pigment

- Developed a pressure sensitive brush system supporting tablet input with stroke interpolation, masking, and batched GPU uploads to minimize CPU–GPU transfer overhead
- Modeled optical color mixing using Beer–Lambert absorption for realistic glazing and wet in wet behavior, including pigment lifting with wet before dry priority
- Wrote a research style technical paper describing the simulation design and planned evaluation; *links*: GitHub | project page | paper (PDF) | demo video

Experience

Computer Science Tutor — Berkeley City College, Berkeley, CA — Sep 2025 – May 2026

- Tutor students across all computer science courses through the Learning Resources Center
- Give short, targeted mini lectures on core CS concepts (data structures & algorithms, x86 assembly, computer organization)
- Guide students on assignments by focusing on problem decomposition, debugging strategies, and conceptual understanding

Math Tutor — Laney College Math Lab, Oakland, CA — Aug 2022 – May 2023

- Provide in person and online tutoring for students across the Peralta district, from Algebra through Calculus III
- Support 3–4 tutors in a shared lab environment, helping students with both conceptual understanding and homework
- Adapt explanations to diverse backgrounds and levels of preparation

Embedded Computer Science Tutor — Berkeley City College, Berkeley, CA — Aug 2022 – May 2023

- Embedded tutor for an introductory computer science course with Prof. Benjamin Allen
- Help students develop algorithmic thinking rather than giving direct solutions, guide them through debugging and problem decomposition
- Provide both in class and outside of class support for students struggling with core CS concepts

Earlier Professional Experience — Korea and China — 2012 – 2019

- Worked in industrial engineering, semiconductor equipment, technical sales, and logistics roles before returning to school, including clean-room equipment installation, equipment repair, technical support, and customer-facing coordination

Main Art Instructor — Mokwoo Art Institution, Korea — Mar 2009 – Dec 2010

- Led a class of ∼50 high school students preparing for competitive entrance exams to art universities
- Designed and ran 5 hour studio “demo tests” simulating real exam conditions, provided detailed feedback on drawing and composition
- Balanced technical instruction with emotional support, building student confidence and improving outcomes

Skills

Programming: Python, C/C++, Java, GDScript, GLSL, Bash
Libraries & Tools: PyTorch, pandas, NumPy, SQL, Godot Engine 4, Git, Linux
Graphics & Interactive Systems: GPU programming, compute shaders, real-time physics simulation, optical color mixing, interactive tool design
Machine Learning & Data: LSTM, feedforward networks, PCA, SVD, data preprocessing and analysis
Systems: x86, RISC-V
Art & Design: Watercolor, oil, acrylic, graphite; visual composition and critique